

Practical Public-Key-Only Forgery Against the Submitted VDOO-128 Implementation

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Abstract

We give a key-only existential forgery attack against the VDOO-128 reference implementation submitted to the Next-generation Commercial Cryptographic Algorithms Program. The attack is mathematical: it uses only the public quadratic map, makes no signing queries, and does not use the independently reported random-number-generator failure. The central-map generator retains, in each oil equation, products with only the oil variable having the same index as that equation. The submitted map is therefore a hidden one-variable-per-equation triangular system rather than the dense UOV layers defined in the specification.

We recover the two hidden oil spaces through rank-one polar derivatives. A small projective multipencil search finds all 40 lines of the second oil layer. After quotienting by their input and output spans, a five-evaluation meet-in-the-middle multipencil finds the 14 lines of the first oil layer. A low-essential-rank filtration then recovers an equivalent flag for the 16 diagonal equations. This equivalent structure suffices to invert the public map: first invert the 16-layer quotient, then lift through two linear systems.

For the claimed 128-bit parameter set over $\mathbb{F}_{16}$, an unoptimized single-threaded Python/NumPy implementation completes a fresh public-key attack and produces an 85-byte signature accepted by the submitted verifier in under twelve minutes on a 2015 desktop-class Intel Core i5-6500. A second experiment uses an independently seeded key to separate the structural attack from the submission’s deterministic-RNG defect. The result applies to the submitted coefficient mask. It does not establish an attack on the dense oil layers in the normative VDOO specification.

1 Introduction

VDOO is a multivariate signature design whose central map combines a diagonal layer with two Unbalanced Oil-and-Vinegar (UOV) layers [1, 2]. For the NGCC VDOO-128 parameter set, the public key is a homogeneous quadratic map

$$P : \mathbb{F}_{16}^{137} \longrightarrow \mathbb{F}_{16}^{70},$$

hidden by invertible linear input and output transformations. Its parameter tuple is

$$(q, v, d, o_1, o_2) = (16, 67, 16, 14, 40).$$

The submission claims 128-bit classical security and reports an 85-byte signature.

The reference implementation does not generate the oil layers described by the specification. In a dense UOV layer, every output polynomial contains vinegar–oil products with

every oil variable. The submitted coefficient filter instead gives output j products with only oil variable j . Thus the last 40 outputs have the form

$$h_j(u, z) = a_j(u) + z_j \ell_j(u), \quad 1 \leq j \leq 40,$$

and, after removing this layer, the preceding 14 outputs have the same form. The oil linear systems have silently become diagonal systems. More significantly for cryptanalysis, every coordinate oil direction has a rank-one public derivative.

This yields a practical attack with the following stages.

1. Recover the 40 hidden second-oil coordinate lines from rank-one derivatives and form their input and output spans O_2, Q_2 .
2. Pass to the induced map $\mathbb{F}_{16}^{137}/O_2 \rightarrow \mathbb{F}_{16}^{70}/Q_2$ and recover the 14 first-oil lines, giving O_1/O_2 and Q_1/Q_2 .
3. Quotient once more. The remaining map $\mathbb{F}_{16}^{83} \rightarrow \mathbb{F}_{16}^{16}$ is a hidden stepwise triangular map. Recover its nested output and input flags from the essential ranks and polar radicals of public quadratic forms.
4. Invert the 16-layer quotient and lift the preimage through O_1/O_2 and O_2 by linear algebra.

The attack belongs to the broad family of attacks on stepwise triangular systems that use chains of kernels [5]. Its first two stages are specialized to the submitted VDOO mask: projective multipencils turn the rank-one derivative signal into two small exhaustive searches. The attack recovers only the equivalent structure needed for inversion; it never needs the submitted secret key.

The public NGCC report already records unrelated implementation failures, including an uninitialized deterministic generator, prehash limits, and secret-dependent branches [4]. Our experiments explicitly initialize the generator for an independent key and discard the secret key before attacking. Consequently, fixing the RNG does not remove this attack.

Scope. We analyze the VDOO-128 reference source at NGCC harness commit `32f381e`. The normative formulas in the VDOO specification contain the full vinegar–oil sums. Our result is a complete forgery of the submitted implementation distribution, not a claim against that dense specified distribution.

2 Preliminaries

Let $K = \mathbb{F}_{16}$. All maps in this paper are over K . For a homogeneous quadratic map $P : K^n \rightarrow K^m$, define its polar map and the derivative at x by

$$\begin{aligned} B_P(x, y) &:= P(x + y) - P(x) - P(y), \\ J_x(y) &:= B_P(x, y). \end{aligned}$$

In characteristic two, subtraction equals addition. The map B_P is bilinear and alternating in each scalar quadratic component, while J_x is a linear map $K^n \rightarrow K^m$.

Suppose $P = S \circ F \circ T$ for invertible linear maps T and S . Then

$$J_x^P = S J_{Tx}^F T,$$

so rank J_x is invariant under the secret hiding transformations. The image line of a rank-one derivative is transformed by S , and its input line is transformed by T^{-1} . These are therefore public invariants.

2.1 Quadratic rank in characteristic two

A scalar quadratic form f is represented by its alternating polar matrix A_f together with its square-term coefficients. We use the *essential rank* $\text{erank}(f)$, the least number of variables needed to express f after an invertible linear change of variables. It is computable without solving a polynomial system:

$$\text{erank}(f) = \begin{cases} \text{rank } A_f, & f|_{\text{rad}(A_f)} = 0, \\ \text{rank } A_f + 1, & f|_{\text{rad}(A_f)} \neq 0. \end{cases}$$

Here $\text{rad}(A_f) = \ker A_f$. Evaluation of f on a basis of the radical decides which case holds. This correction is necessary over characteristic two; polar rank alone loses square terms.

2.2 Compatible quotients

The following elementary observation is used twice.

Lemma 1 (Quadratic quotient). *Let $Z \subseteq K^n$ and $R \subseteq K^m$ satisfy*

$$P(Z) = 0, \quad B_P(K^n, Z) \subseteq R.$$

Then $\bar{P}(x + Z) = P(x) + R$ is a well-defined quadratic map $K^n/Z \rightarrow K^m/R$.

Proof. For $z \in Z$, $P(x + z) = P(x) + P(z) + B_P(x, z) \equiv P(x) \pmod{R}$. $\square$

Once bases for Z and R are public, the quotient is computed with ordinary Gaussian elimination: select a complement of Z , and left-compose the outputs with a basis of the annihilator of R .

3 The submitted coefficient-mask defect

Write the central variables as

$$(u, d, y, z) \in K^{67} \times K^{16} \times K^{14} \times K^{40}.$$

The specification defines each first-oil output with all products $x_i y_j$, for $1 \leq i \leq 83$ and $1 \leq j \leq 14$, and each second-oil output with all products $x_i z_j$, for $1 \leq i \leq 97$ and $1 \leq j \leq 40$ [2].

In the submitted source, the admissibility predicate for first-oil output r sets

$$\text{oil_var_idx} = 83 + r$$

and accepts a vinegar–oil monomial only when its second index is exactly `oil_var_idx`. The second-oil predicate repeats the same construction with prefix length 97. Algebraically, the generated layers are

$$g_j(w, y) = q_j(w) + y_j \lambda_j(w), \quad w \in K^{83}, \quad 1 \leq j \leq 14, \quad (1)$$

$$h_k(a, z) = r_k(a) + z_k \mu_k(a), \quad a \in K^{97}, \quad 1 \leq k \leq 40. \quad (2)$$

All coefficients in $q_j, r_k, \lambda_j, \mu_k$ are sampled before the forbidden monomials are zeroed. With overwhelming probability, the displayed linear forms are nonzero and independent within each layer.

The diagonal layer already has one new variable per equation. Its i th output is

$$f_i(u, d_1, \dots, d_i) = a_i(u, d_1, \dots, d_{i-1}) + d_i b_i(u, d_1, \dots, d_{i-1}), \quad 1 \leq i \leq 16. \quad (3)$$

Equations (1)–(3) show that the entire submitted central map is a 70-step triangular system with one new variable per output, although the 14- and 40-equation blocks omit permitted dependencies on earlier variables in their own blocks.

3.1 Rank-one derivative lines

Let e_k be the k th coordinate vector of the last oil space. Equation (2) gives

$$J_{e_k}^F(a', z') = (0, \dots, 0, \mu_k(a'), 0, \dots, 0).$$

Thus $J_{e_k}^F$ has rank one whenever $\mu_k \neq 0$, and its image is the k th second-oil output line. Also $F(e_k) = 0$.

Proposition 2 (Public oil-line signal). *Let $x_k = T^{-1}e_k$ and let q_k be the image under S of the k th second-oil output line. Then*

$$P(x_k) = 0, \quad \text{rank } J_{x_k}^P = 1, \quad \text{im } J_{x_k}^P = q_k.$$

Consequently the projective lines $\langle x_k \rangle$ are detectable from the public key, their span is the hidden space O_2 , and the span of their derivative images is Q_2 .

Inside O_2 , a vector supported on s coordinate lines has derivative rank s provided the corresponding μ_k are independent. Hence the rank-one points in $\mathbb{P}(O_2)$ are precisely the 40 coordinate lines with overwhelming probability. The same argument applies to the 14 first-oil lines after quotienting by (O_2, Q_2) .

4 Projective multipencil recovery

Directly enumerating $\mathbb{P}(K^{137})$ is impossible. We instead convert the rank-one condition into a small search over output ratios.

Choose evaluation points $a_1, \dots, a_t \in K^n$ and define public linear maps

$$L_i : K^n \longrightarrow K^m, \quad L_i(x) = B_P(x, a_i).$$

If $\text{rank } J_x = 1$, then all $L_i(x)$ lie on one output line. Therefore there are $u \in K^m$ and a projective label $c = (c_1 : \dots : c_t) \in \mathbb{P}^{t-1}(K)$ such that $L_i(x) = c_i u$. Choose a pivot p with $c_p \neq 0$. Then x lies in the kernel of

$$A_c = \begin{bmatrix} c_p L_1 - c_1 L_p \\ \vdots \\ c_p L_t - c_t L_p \end{bmatrix}_{i \neq p}. \quad (4)$$

For each label c , we compute $\ker A_c$ and retain only vectors whose full public derivative has rank one. A final check requires $P(x) = 0$.

Lemma 3 (Multipencil completeness). *Let $x \neq 0$ satisfy $\text{rank } J_x = 1$. If not all $J_x(a_i)$ vanish, then the projective label $c = (J_x(a_1) : \dots : J_x(a_t))$ is well defined after identifying the one-dimensional image of J_x with K , and $x \in \ker A_c$.*

Proof. Write $J_x(y) = \ell_x(y)u$ for a nonzero u . Take $c_i = \ell_x(a_i)$. Every row block in (4) annihilates x because $c_p L_i(x) - c_i L_p(x) = c_p c_i u - c_i c_p u = 0$. $\square$

For random evaluations, the excluded event has probability q^{-t} for a fixed rank-one line. If a run finds fewer than the expected number of independent lines, it is simply repeated with fresh evaluations. Every output is checked by the full derivative, so this is a Las Vegas procedure.

4.1 Second oil layer

At the first stage $(n, m) = (137, 70)$. Three evaluations suffice because $2m = 140 \geq 137$. There are only

$$|\mathbb{P}^2(\mathbb{F}_{16})| = 16^2 + 16 + 1 = 273$$

labels. For each label we reduce a 140×137 matrix. The verified 40 lines span O_2 ; one nonzero column of each rank-one derivative supplies its image line, and these span Q_2 . Public checks give

$$\dim O_2 = \dim Q_2 = 40, \quad P(O_2) = 0, \quad \text{quad}B_P(K^{137}, O_2) \subseteq Q_2.$$

The quotient therefore has dimensions

$$\overline{P}_2 : K^{97} \longrightarrow K^{30}.$$

4.2 First oil layer and the $3 + 2$ split

For $(n, m) = (97, 30)$, five evaluations are needed because $3m < 97 \leq 4m$. A direct projective search would have $(16^5 - 1)/15 = 69,905$ labels and reduce a 120×97 matrix for each one. We use a $3 + 2$ split.

First enumerate the 273 projective labels for the first three evaluations. The first two proportionality relations leave a typical kernel K_c of dimension

$$97 - 2 \cdot 30 = 37.$$

For every such K_c , enumerate the two remaining affine ratios $(a, b) \in K^2$. The last two relations are then a 60×37 system on K_c . This requires

$$273 \cdot 16^2 = 69,888$$

small reductions. The 17 projective labels whose first three entries vanish are intentionally omitted. A desired line is omitted only when three random evaluations all lie in the kernel of its derivative, an event of probability 16^{-3} . Restarting if fewer than 14 lines are found makes the algorithm Las Vegas.

The resulting 14 verified lines span O_1/O_2 and their derivative images span Q_1/Q_2 . Quotienting again produces

$$P_D : K^{83} \longrightarrow K^{16}.$$

5 Recovering the diagonal flag

The quotient P_D is linearly equivalent to the 16 equations in (3). Let

$$\{0\} = J_0 \subset J_1 \subset \cdots \subset J_{16} = K^{16}$$

be the hidden output flag, where J_ℓ is the span of the first ℓ diagonal forms. Every form in J_ℓ uses at most $v + \ell = 67 + \ell$ essential variables, so

$$J_\ell \subseteq \mathcal{R}_{67+\ell} := \{c \in K^{16} : \text{erank}(c \cdot P_D) \leq 67 + \ell\}. \quad (5)$$

For random coefficients, J_ℓ is the unique codimension-one linear component of $J_{\ell+1} \cap \mathcal{R}_{67+\ell}$. Individual points outside the hidden hyperplane can be accidentally rank deficient, so accepting a point from one rank test is unsafe. We instead grow a linear space while

checking projective line closure. A candidate x is admitted only when, for random confirmed z and every $\lambda \in K^*$,

$$x + \lambda z \in \mathcal{R}_{67+\ell}.$$

After a hyperplane has dimension ℓ , 32 additional random combinations are tested. If a check fails, that descent is restarted. Beginning with $J_{16} = K^{16}$ and descending to J_1 recovers the complete output flag.

Choose ordered forms $p_1, \dots, p_{16}$ such that $\text{span}(p_1, \dots, p_\ell) = J_\ell$. Their common polar radicals reveal the input-tail flag

$$K_\ell = \bigcap_{i=1}^{\ell} \text{rad}(B_{p_i}), \quad \dim K_\ell = 16 - \ell. \quad (6)$$

The one-dimensional quotients $K_{\ell-1}/K_\ell$ give directions for the ℓ th new variable. Both the dimensions in (6) and all rank bounds in (5) are public, efficiently checkable certificates of the recovered equivalent flag.

6 Public-map inversion and forgery

Let $t \in K^{70}$ be an arbitrary target. Project it successively to the 30- and 16-dimensional quotient codomains. We invert in the opposite order.

6.1 Inverting the diagonal quotient

The first ordered form ignores the 15-dimensional tail K_1 . Choose a random point in a fixed 68-dimensional complement until $p_1(x) = t_1$; this takes about 16 trials. For $\ell = 2, \dots, 16$, choose a representative $w_\ell \in K_{\ell-1} \setminus K_\ell$ and enumerate $a \in \mathbb{F}_{16}$ until

$$p_\ell(x + aw_\ell) = t_\ell.$$

Earlier equations remain unchanged because w_ℓ lies in their common polar radical and in their zero tail. If a layer has no solution, restart from a new first-layer point. The final public evaluation certifies the quotient preimage.

6.2 Linear lifts

Let x be a representative of the diagonal-quotient solution in K^{97} . For $o \in O_1/O_2$, the quotient map satisfies

$$\overline{P}_2(x + o) = \overline{P}_2(x) + B_{\overline{P}_2}(x, o),$$

because the recovered oil space is a zero subspace. The residual lies in Q_1/Q_2 , so its cancellation is a linear system in the 14 coordinates of o . If the system is singular or inconsistent, use another diagonal preimage.

Lift the resulting point to K^{137} and repeat the same operation with $o \in O_2$. This time the linear system has 40 unknowns and a residual in Q_2 . A successful solution s is checked by evaluating the full public map and requiring $P(s) = t$.

To forge a signature on a chosen fresh message M , choose a 16-byte salt ρ , compute the submission's target

$$t = H(H(M) \parallel \rho) \in \mathbb{F}_{16}^{70},$$

and apply the inversion above. Pack $s \in \mathbb{F}_{16}^{137}$ in 69 bytes and append ρ . This is an 85-byte signature in the submitted format.

Theorem 4 (Conditional correctness). *Assume that the two multipencil searches recover the stated rank-one line sets, the diagonal rank filtration returns flags satisfying (5)–(6), and the two lift systems have a solution for one sampled quotient preimage. Then the algorithm outputs a signature accepted by the submitted VDOO-128 verifier for the chosen message and salt.*

Proof. The quotient lemma shows that each recovered oil pair defines the induced quadratic map used by the next stage. The tail directions preserve all previous diagonal equations, and exhaustive enumeration in $\mathbb{F}_{16}$ sets the next equation; hence the diagonal output is correct. Each lift uses the identity $P(x + o) = P(x) + B_P(x, o)$ on a zero oil space and solves exactly for the remaining residual. Therefore the final vector satisfies $P(s) = H(H(M) \parallel \rho)$, which is precisely the submitted verification condition. $\square$

All hypotheses are checked from the public key. A failed random choice causes a restart rather than an incorrect output. Final verifier acceptance is an end-to-end certificate independent of the structural interpretation.

7 Complexity

Table 1 gives the fixed search sizes for VDOO-128. The costs are polynomial except for enumerations over very small projective spaces whose sizes are shown explicitly. There is no term near 2^{128} .

Table 1: Main public-key work for submitted VDOO-128.

Stage	Search	Dominant linear algebra
Recover O_2, Q_2	273 labels in $\mathbb{P}^2(\mathbb{F}_{16})$	kernels of 140×137 matrices
Recover $O_1/O_2, Q_1/Q_2$	$273 \cdot 256 = 69,888$ split labels	273 kernels of 60×97 , then kernels of 60×37
Recover diagonal flag	about 5.3×10^3 sampled projective rank tests	essential rank of forms on at most 83 variables
Invert one target	about 16 trials for the first form, plus retries	14- and 40-unknown linear lifts

The projective searches can be parallelized label by label. Our proof-of- concept does not do so. It uses byte lookup tables for $\mathbb{F}_{16}$ arithmetic and vectorized row operations; it does not use compiled finite-field kernels, SIMD, GPUs, or multiple cores. The reported timings should therefore be read as a practical upper bound rather than an optimized record.

8 Experiments

8.1 Environment and methodology

We built the submitted VDOO-128 reference sources from NGCC harness commit `32f381e`. Experiments ran on an Intel Core i5-6500 at 3.20 GHz (four physical cores), with 15 GiB RAM, Python 3.13.13, and NumPy 2.5.3. The attack process is single-threaded.

For the independent-key trial, we initialized the implementation’s private DRNG with the 48-byte seed $00, 01, \dots, 2f$ before key generation. The secret key returned by key generation was discarded. The attack process received only the serialized public key. This bypasses the already reported zero-initialized-RNG behavior and tests a different, independently sampled central map and hiding transformations.

8.2 Results

Table 2: Uncached end-to-end attack on the independently seeded key. “Online” includes quotient inversion, both lifts, signature packing, and verification.

Trial	O_2	O_1	flag/checks	online	total	result
Independent seeded key	25.0 s	380.4 s	201.8 s	78.0 s	693.2 s	accept

For both keys the first multipencil returned exactly 40 projective rank-one lines spanning a 40-dimensional public zero subspace, and their images spanned 40 dimensions. The quotient multipencil returned exactly 14 lines spanning a 14-dimensional zero subspace and 14 output dimensions. The diagonal stage recovered all 15 proper output-flag subspaces and the tail dimensions 15, 14, $\dots$, 0. Every stage was checked using only the public map.

The first complete structural run took 600.9 seconds to invert an arbitrary target. Reusing the public-key structure, a subsequent fresh-message forgery was accepted in 25.5 seconds of online inversion. The independent seeded-key run started without cached structure and ended with acceptance by the same submitted verification function. Peak resident memory remained below the available 15 GiB; the persistent public-key attack state occupies less than one megabyte.

8.3 Accepted independent-key forgery

The independently seeded experiment forged the following message without a signing query:

```
Independent-key public-key-only
structural forgery against submitted VD00-128
```

The submitted verifier returned zero (accept). Flipping one bit of the message or one bit of the signature made it return -1 (reject). The binary artifacts have the following SHA-256 digests:

Object	SHA-256
Public key	f0bf3e7faaa200c14c8b37c12d4a0d7195d2ff6c0ce8081e29ea303a4b3d91c6
Message	31b942f4df86978090308e2bc13a54379dd0333c66d5210831d4ddf49ceeb575
Signature	575c13a9f988e6210c562c2cab861dc13228878c1aa45d7d7357f6188ea5b08c

The 85-byte signature, including its final 16-byte salt, is

```
9f3606a0748596e7bf00a719fd1234dbb562a86fa6f465e377f81d1f6b8f8d63
f28b5577b42cc809d5d76d9fdbb77c7b6533b561a2aef559b986dbc4972c1311
3ba7d2e00f33d253ff531ba22f82951e499884ed23
```

These values are included to make the acceptance claim independently checkable without relying on any attack implementation.

9 Why the attack survives an RNG repair

The public RNG failure allows an even simpler secret-key reproduction attack [4]. It is not used here. The signal exploited in this paper is the public tensor identity

$$\text{rank}(y \mapsto B_P(x_k, y)) = 1$$

for every hidden matching-oil direction. This identity follows from which coefficients are set to zero after random sampling. Reseeding changes the nonzero coefficients and hiding transformations, but it preserves the identity. The independently seeded forgery confirms this distinction.

Likewise, the attack needs no fault, timing trace, malformed key, reused signature randomness, chosen-target oracle, or known signature. Its input is a well-formed public key. Since it forges a fresh chosen message in the key-only setting, it directly violates existential unforgeability and is stronger than an attack requiring chosen-message access.

10 Remediation and design lesson

The immediate repair is to generate the full vinegar–oil block specified for each oil equation. For the first layer, output r must permit

$$0 \leq i < 83, \quad 83 \leq j < 97,$$

for every j , rather than only $j = 83 + r$. For the second layer, it must permit

$$0 \leq i < 97, \quad 97 \leq j < 137$$

for every j . Test vectors and key-size calculations must then be regenerated and the repaired distribution reanalyzed. The RNG and side-channel findings require separate fixes.

Unit tests should test structural distribution properties, not only sign–verify correctness. In particular, for random secret-coordinate oil directions in a dense UOV layer, the derivative image should span many output coordinates. A test that counts the public rank-one derivative locus after a small hidden toy instantiation would have exposed the one-variable mask while ordinary known-answer tests continued to pass.

The broader lesson is that a coefficient filter is part of the cryptographic design. Replacing a dense UOV linear system by per-equation matching columns does more than optimize storage or signing: it changes the isomorphism class of the public quadratic tensor and exposes invariant rank-one points. Hiding linear transformations cannot erase those tensor ranks.

11 Conclusion

The submitted VDOO-128 implementation can be forged from its public key in minutes on an ordinary laptop-class machine. The attack recovers rank-one oil directions, forms two public quotients, reconstructs an equivalent diagonal flag, and inverts a fresh hash target. A signature produced for an independently seeded key is accepted by the submitted verifier. The attack is caused by a sparse coefficient mask that selects only the oil variable matching each output equation. Restoring the dense UOV layers in the normative specification removes this particular rank-one mechanism, but requires new test vectors and fresh cryptanalysis.

A Attack pseudocode

Rank-one multipencil. Given the polar map $B : K^n \times K^n \rightarrow K^m$ and an evaluation count t :

1. sample $a_1, \dots, a_t \leftarrow K^n$ and set $L_i(x) = B(x, a_i)$;
2. enumerate $c \in \mathbb{P}^{t-1}(K)$, choose a nonzero pivot c_p , and form A_c as in (4);

3. enumerate the projective points of each small $\ker A_c$;
4. retain x precisely when $\text{rank } J_x = 1$ and $P(x) = 0$;
5. normalize projective representatives and remove duplicates;
6. restart with new a_i if the verified span has the wrong dimension.

For the $3 + 2$ variant, enumerate a projective label for L_1, L_2, L_3 , compute the kernel of the first two relations, restrict L_p, L_4, L_5 to that kernel, and enumerate the final two affine ratios.

Full inversion.

1. use the three-evaluation multipencil to recover O_2, Q_2 ;
2. construct $P_2 : K^{137}/O_2 \rightarrow K^{70}/Q_2$;
3. use the five-evaluation split multipencil to recover $O_1/O_2, Q_1/Q_2$;
4. construct $P_D : K^{83} \rightarrow K^{16}$;
5. recover $J_{15}, \dots, J_1$ by descending essential-rank hyperplanes and derive the tail radicals $K_1, \dots, K_{16}$;
6. invert the projected target through the 16 one-dimensional layers;
7. solve the 14-variable first-oil lift and the 40-variable second-oil lift, restarting the diagonal inversion after an inconsistent lift;
8. require full public evaluation $P(s) = t$ before returning s .

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